

BAYANTOOROI, Mongolia

WMO: 443250

Lat: 44.924N

Lon: 96.753E

Elev: 7156

StdP: 11.27

Time Zone: 7.00 (E07)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.9	-21.4	-32.0	1.2	-23.4	-27.0	1.6	-15.7	34.8	8.0	23.1	11.4	2.8	90	0.591

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.6	94.7	64.8	91.9	62.4	89.2	61.1	70.2	87.3	67.7	83.6	65.5	81.6	8.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.6	123.8	78.9	63.0	112.9	75.9	60.3	102.3	75.9	39.3	88.0	36.8	83.6	34.7	81.7	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 33.8	26.6	22.5	DB	-24.6	99.0	8.9	2.3	-30.9	100.6	-36.1	101.9	-41.1	103.2	-47.5	104.9	(4)
(5)			WB	-25.6	70.6	8.6	4.9	-31.7	74.2	-36.8	77.1	-41.6	79.8	-47.8	83.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.2	0.4	10.7	32.9	50.4	61.1	73.0	77.3	73.9	61.2	44.5	24.3	6.6		
	DBStd	27.92	11.29	12.68	11.28	9.20	7.51	6.04	5.48	5.65	7.02	7.92	9.99	9.90		
	HDD50	5612	1539	1101	531	104	9	0	0	0	7	204	770	1347		
	HDD65	8954	2004	1521	994	439	162	11	1	4	151	635	1220	1812		
	CDD50	3130	0	0	2	117	354	691	848	740	344	34	0	0		
	CDD65	997	0	0	0	3	42	252	383	279	38	0	0	0		
	CDH74	11378	0	0	0	45	550	2863	4504	3010	405	1	0	0		
	CDH80	4722	0	0	0	5	134	1158	2123	1227	75	0	0	0		
	WSAvg	7.0	3.9	5.2	7.9	9.7	10.3	8.5	8.0	7.8	6.8	6.3	5.2	4.4		
	PrecAvg	1.9	0.0	0.0	0.0	0.1	0.0	0.2	0.7	0.5	0.2	0.0	0.0	0.0		
	PrecMax	5.4	0.3	0.2	0.2	0.4	0.3	1.0	2.2	2.2	0.8	0.4	0.2	0.2		
(15) Precipitation	PrecMin	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	1.1	0.1	0.0	0.0	0.1	0.1	0.3	0.6	0.6	0.2	0.1	0.0	0.0		
	DB	29.5	44.9	63.4	79.5	88.6	95.9	99.4	95.8	86.2	71.1	53.5	32.9			
	MCWB	22.4	32.8	42.5	53.3	56.1	64.4	66.5	64.1	55.8	51.9	39.5	24.7			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	23.6	38.0	59.3	74.1	84.2	92.6	96.1	92.2	81.7	66.4	47.9	27.5			
	2% MCWB	18.1	28.1	40.5	50.2	57.1	62.7	66.5	61.2	54.9	47.9	36.9	21.5			
	5% DB	19.4	32.9	55.7	70.9	80.4	89.6	93.1	89.8	78.7	62.8	43.0	23.8			
	5% MCWB	14.9	24.8	38.7	48.8	55.3	61.0	64.5	60.7	54.6	46.1	33.7	18.7			
	10% DB	15.1	28.7	51.3	67.1	76.7	86.5	90.2	87.0	75.3	59.2	38.7	20.0			
	10% MCWB	11.5	22.2	36.3	46.8	54.1	60.4	62.4	60.4	53.1	44.2	30.7	15.8			
	0.4% WB	22.7	32.9	44.5	55.9	63.5	70.4	75.1	70.9	61.5	54.8	40.9	25.4			
	0.4% MCDB	29.0	44.1	59.8	74.1	79.2	87.6	92.0	84.6	74.5	67.2	50.5	31.9			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	18.3	28.3	41.5	52.8	60.2	67.6	72.0	68.0	59.1	50.9	37.4	21.9			
	2% MCDB	23.5	37.3	57.2	70.8	78.9	84.8	90.0	83.1	72.8	63.6	47.4	27.2			
	5% WB	14.9	25.3	39.2	50.1	57.8	65.6	69.3	65.8	57.1	47.7	34.2	18.9			
	5% MCDB	19.2	32.1	54.8	68.0	76.1	83.1	87.0	80.6	73.2	60.0	42.8	23.7			
	10% WB	11.7	22.3	36.8	47.8	55.4	63.8	66.5	63.7	55.1	44.9	30.8	15.9			
	10% MCDB	15.0	28.7	51.0	65.4	73.8	80.5	82.2	80.2	70.9	57.6	38.3	20.0			
	MDBR	21.2	24.1	26.4	24.9	23.6	22.4	21.6	21.7	23.6	24.8	22.2	20.1			
	5% DB	24.4	27.2	28.5	27.9	28.3	26.1	24.2	23.9	26.0	27.8	26.2	22.7			
(36) Mean Daily Temperature Range	MCWBR	20.0	19.7	16.6	15.2	13.9	11.5	9.8	9.8	11.6	15.7	18.0	18.0			
	MCDBR	24.1	26.5	26.9	26.7	24.5	22.6	21.2	20.2	23.7	25.4	26.0	22.7			
	MCWBR	19.9	19.5	16.1	15.9	14.7	11.6	11.2	10.5	12.5	16.1	18.1	18.0			
	MCDBR	24.1	26.5	26.9	26.7	24.5	22.6	21.2	20.2	23.7	25.4	26.0	22.7			
(40) Clear-Sky Solar Irradiance	taub	0.232	0.260	0.295	0.330	0.339	0.355	0.351	0.330	0.300	0.274	0.247	0.232			
	taud	2.435	2.371	2.287	2.246	2.280	2.308	2.356	2.403	2.450	2.456	2.456	2.428			
	Ebn at Noon	291	298	298	295	294	289	289	292	294	289	281	279			
	Edh at Noon	22	28	36	41	41	40	38	35	30	26	21	20			
(44) All-Sky Solar Radiation	RadAvg	646	1000	1505	1964	2294	2266	2183	1966	1677	1199	744	563			
	RadStd	61	64	93	91	124	129	92	100	60	46	58	49			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	-5.90	N/A	-4.07	-4.34	N/A	N/A	N/A	N/A

Nomenclature: See separate page